

Research on Face Recognition Technology Based on PCA and SVM

Yan Wang

School of Information Engineering
Wuhan Business University
Wuhan, China
wangyanwsr@163.com

Qinglin Wu

School of Information Engineering
Wuhan Business University
Wuhan, China
254275383@qq.com

Abstract—Face recognition is a hot topic in the field of artificial intelligence and pattern recognition, which is mainly conducive to machine learning algorithm to judge whether face samples belong to a certain category. Based on the discussion and analysis of PCA and SVM technology, this paper expounds the basic principle and process of face recognition, and puts forward face recognition method based on PCA and SVM. Finally, this method is used to recognize LFW (labeled faces in the wild) face data set, and the parameter adjuster named gridsearchcv is used to adjust the parameters to further improve the precision and recall of face recognition. The experimental results show that the face recognition technology based on PCA and SVM not only reduces the dimension of face data, but also increases the recognition accuracy, which is a feasible face recognition method.

Keywords—face recognition, PCA, SVM, characteristic vector

I. INTRODUCTION

With the development of artificial intelligence technology, face recognition technology has developed very rapidly in recent years. It is widely used in different fields of production and life. It has attracted extensive attention from industry and academia. It has become an important function of various application products in modern society, which is conducive to promoting the development of industry. Face recognition technology is based on human facial features for identity identification. It has important theoretical research value and application value in industrial production, education and teaching, traffic safety and health care. Feature extraction is to extract effective features from existing face features and combine them into a new feature subset according to the needs of face recognition. In the process of face recognition, firstly, check whether the image has face information from the collected image. If there is face information, select the face part from the image and extract the face feature information, such as the relative relationship between the positions of eyes, nose and eyes, and then select an appropriate classifier to model the effective feature information. Aiming at the problem that the pixel dimension of face image is too high, which is not conducive to machine storage and processing, this paper proposes a face recognition technology method based on principal component analysis and support vector machine[1-2]. Firstly, the face image is expressed as a column vector, and the required recognition information is retained after PCA transformation by principal component analysis. Then the face

image is recognized by SVM, and the effectiveness of the method is tested by python programming language.

The paper is organized as follows: Section 2 presents the principle of principal component analysis, and section 3 discusses the method of support vector machine, section 4 analyses the basic principle of face recognition, and the face recognition process based on PCA and SVM is proposed in section 5. Finally, it is verified by the data set LFW (labeled faces in the wild) face database.

II. PRINCIPAL COMPONENT ANALYSIS

Principal component analysis can be regarded as a neural network with only one layer. Its main idea is to reduce data redundancy. In the case of dimension reduction, it can also achieve the purpose of data processing. A small number of dimensions are used to represent the original data, which is convenient for better processing and storage of data, and convenient for data visualization [3-4]. The process of dimensionality reduction is essentially a process of linear transformation. Its core idea is to project the data along the direction of maximum variance to retain the characteristics of the original data as much as possible.

The main advantages of PCA algorithm are: the principal components are orthogonal, and the data after dimensionality reduction have no influence on each other. The calculation method is simple, and the main operation is eigenvalue decomposition, which is easy to implement. The main disadvantages of PCA algorithm are: the meaning of each feature dimension of principal component has a certain fuzziness, which is not as strong as the interpretation of the features of the original sample. The loss of non principal components may affect the subsequent data processing. Assuming that for M samples with n features $X=\{x_1, x_2, \dots, x_m\}$, the dimension should be reduced to dimension n, the main steps of applying principal component analysis are:

(1) Centralize all samples, that is, subtract the respective average value from each feature.

$$x_i = x_i - \frac{1}{m} \sum_{j=1}^m x_j \quad (1)$$

(2) Calculate the covariance matrix of the sample, When the covariance is 0, it means that the two fields are completely irrelevant, which can maximize the information represented by the feature. In order to make the covariance 0, when we select

the second basis, we can only select in the direction that the first basis is orthogonal, so as to achieve the purpose that the two direction vectors are orthogonal

$$C = \frac{1}{m} XX^T \quad (2)$$

(3) The eigenvalues and corresponding eigenvectors of the covariance matrix are calculated. The elements on the diagonal of the covariance matrix are the variances of the fields respectively, while the other elements are the covariance of the fields.

(4) Sort the eigenvalues from large to small, select the largest K of them, and then take the corresponding K eigenvectors as row vectors to form the eigenvector matrix P.

(5) According to the calculation of change matrix $Y = PX$, multiply the P matrix by the original data matrix X to obtain the k-dimensional data matrix Y after dimensionality reduction.

III. SUPPORT VECTOR MACHINE

Support vector machine was proposed by Vladimir N. Vapnik and Alexey Ya. Chervonenkis in 1963. It was originally a binary classification algorithm. After evolution, it also supports multi classification problems and can also be applied to regression problems[5-6].

(1) Linear separable SVM algorithm

Support vector machine learning needs to achieve the following two goals: (1) correctly divide the training data set as much as possible; (2) Find the hyperplane with the largest geometric interval, so that the model has strong generalization ability. Generally speaking, there are infinite separation planes for linearly separable data sets, but the separable hyperplane with the largest geometric spacing is unique. Finding the separable hyperplane with the largest geometric interval is the optimal solution to be found by SVM. Fig. 1 shows a linearly separable support vector machine.

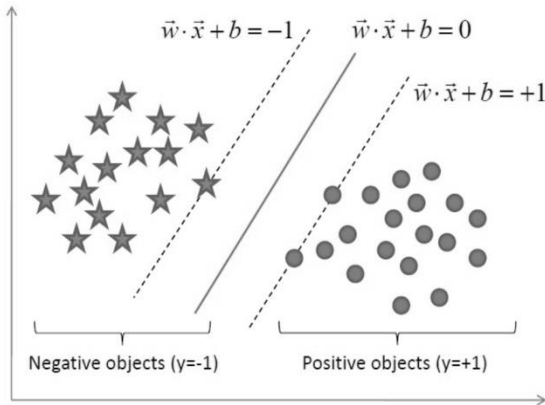


Fig. 1. Linear separable SVM

The maximum segmentation hyperplane problem of SVM model can be expressed as the following constrained optimization problem:

$$\min \frac{1}{2} ||w||^2 \quad (3)$$

$$\text{s. t. } y_i(w \cdot x_i + b) \geq 1, i = 1, 2, \dots, n \quad (4)$$

This is a programming problem with inequality constraints. In order to simplify the processing of data, it is usually considered that the training set data is linearly separable, but in fact, most of the collected data sets are linearly non separable. Therefore, the data result obtained by the above method is not ideal. In order to solve this problem, the concept of "soft interval" is introduced, that is, some points are allowed to not meet the constraints, which increases the generalization of model classification.

(2) Nonlinear SVM algorithm

In practical problems, the classification problem is often nonlinear, so the optimal classification surface we need is also nonlinear, which increases the complexity of machine learning and is prone to over fitting. If the input data set is completely linear and inseparable, the method of changing low dimension to high dimension can be used to solve the nonlinear separability problem, that is, mapping the data of two-dimensional plane into three-dimensional space, and learning linear support vector machine in three-dimensional space. At this time, it is difficult to specify a clear linear transformation. Considering that the objective function and classification decision function of linear support vector machine only involve the inner product between instances, the kernel function replaces the inner product to solve this problem.

Based on the above considerations, support vector machine mainly solves the nonlinear classification problem through kernel function. Kernel function can effectively map samples to high-dimensional feature space, make it linearly separable, and better solve the problem of linear non separability. Kernel function is the main method of support vector machine to solve nonlinear separable problems. It is widely used in different classification problems and has obvious effect.

The commonly used kernel functions include linear kernel, polynomial kernel, Gaussian kernel and sigmoid kernel, the expressions are as follows[7-8]:

(1) linear kernel is actually a linear separable SVM, and the expression is: $K(x, z) = x \cdot z$;

(2) polynomial kernel is one of the commonly used kernel functions of linear non separable SVM, and the expression is: $K(x, z) = (\gamma x \cdot z + r)^d$;

(3) Gaussian kernel, also known as radial basis function (RBF) in SVM, is the most mainstream kernel function of nonlinear classification SVM. It is the default kernel function of libsvm. The expression is $K(x, z) = \exp(-\gamma ||x - z||^2)$;

(4) sigmoid kernel function sigmoid kernel function is also one of the commonly used kernel functions of linear non separable SVM. The expression is: $K(x, z) = \tanh(\gamma x \cdot z + r)$

IV. BASIC FRAMEWORK OF FACE RECOGNITION

The process of face recognition is divided into four parts: image acquisition, face detection, feature extraction and face classification and recognition. Its main functions are as follows [9-10]:

1 image acquisition: face image acquisition is mainly obtained through web crawler acquisition or camera lens

acquisition. The web crawler can get more face images in a short time. The camera lens collects pictures relatively slowly. When the user is within the shooting range of the acquisition device, the acquisition device will automatically search and capture the user's face image.

2 Face detection is mainly about whether there is a face in the input face image. If it includes a face, determine the position and size of the face. Face images contain rich pattern features, such as histogram features, color features, template features and structural features. Face detection is to pick out the useful information and use these features to realize face detection.

3 Face feature extraction, also known as face representation, is the process of feature modeling of human face. The main task is to select the features to distinguish different individual differences based on image preprocessing such as face detection, location and normalization. Face feature extraction methods are divided into knowledge-based representation methods and statistics based representation methods.

Face classification and recognition is the last step of the whole face recognition system. Its main task is to select an appropriate classifier to identify the face identity according to the face feature extraction results, judge the category of the face to be tested according to the similarity of the identification, and output the matching results when the similarity exceeds this threshold. Fig. 1 below shows the basic structure of the model:



Fig. 2. Face recognition process

V. FACE RECOGNITION PROCESS BASED ON PCA AND SVM

Generally, the pixel dimensions of face images are relatively high, and there is a large correlation between the pixels of each dimension of the original image. Therefore, it is not necessary to store all the pixels of the image in the process of face recognition. The main components are selected by principal component analysis (PCA), and the low-dimensional feature vector obtained after dimensionality reduction is used as the feature of face recognition and classification. The reduced dimension data can meet the requirements of face recognition and classification, and facilitate the storage and processing of face image data. The process of face recognition based on PCA and SVM is as follows :

(1) Preliminary pretreatment

The steps of face image preprocessing are geometric normalization, noise processing, gray processing and histogram equalization. Geometric normalization is to unify the geometric information of each input image, including the position, angle, size and other characteristic information of face image. Noise processing mainly eliminates the signal that affects the image quality, reduces the impact of noise on image recognition, and improves the accuracy of face recognition. Gray processing realizes the conversion of color image into black-and-white color image, which makes the face recognition system more convenient for image processing and storage. The purpose of

image histogram equalization is to enhance the contrast of the image, make its distribution more uniform, and reduce the negative impact of illumination change on face recognition.

(2) PCA dimensionality reduction

In the face recognition system, the face image is mainly represented by matrix. The face image is high-dimensional data, which is not conducive to the face recognition algorithm processing of the face recognition system. PCA algorithm is used to reduce the dimension of the extracted face features. On the premise that the face data retains most of the information of the original image data, the original image is transformed from high-dimensional data to low-dimensional data. By changing the feature matrix of face image through PCA, the data dimension is reduced, and the feature face of each face is obtained. Dimension reduction processing effectively retains most of the important features of face images, only loses some unimportant feature information, and does not affect the accuracy of face image recognition.

(3) SVM face classification and recognition

After obtaining the feature face data after PCA dimensionality reduction, the feature face is classified and recognized by SVM classification method. When using SVM to classify and recognize face, two problems need to be considered: (1) the selection of kernel function. Through the kernel function, the feature vector can be easily mapped to a higher dimensional space. In the new dimensional space, the original linear non separable data becomes linearly separable. (2) Selection of parameters of kernel function. Parameter C indicates the degree of punishment for classification errors. If the value of C is large, it indicates that the classifier has high requirements for accuracy, which is easy to cause over fitting; The C value is small, and the accuracy of the classifier is low, which is easy to cause the phenomenon of under fitting. The feature face data reduced by PCA is transmitted to SVM classifier, which can classify the feature face and achieve the goal of face recognition.

VI. FACE RECOGNITION EXAMPLE

In this experiment, the data set LFW (labeled faces in the wild) face database is compiled by the Computer Vision Laboratory of Amherst University of Massachusetts. It is mainly used to study the problem of face recognition under unrestricted conditions and test the accuracy of face recognition. In order to better learn face features, this experiment selects more than 50 avatars on the face. After screening, the data set includes 1560 photos with length and width of 50 pixels and 37 pixels respectively.

Step 1: use SVM algorithm to recognize the face, use RBF kernel function and the class weight parameter is balanced. The experimental results are as follows:

	precision	recall	f1-score	support
Ariel Sharon	0.70	0.67	0.68	21
Colin Powell	0.76	0.82	0.79	57
Donald Rumsfeld	0.67	0.73	0.70	33
George W Bush	0.83	0.81	0.82	111
Gerhard Schroeder	0.65	0.45	0.53	29
Hugo Chavez	0.57	0.76	0.65	21
Jacques Chirac	0.42	0.50	0.45	10
Jean Chretien	0.83	0.71	0.77	14
John Ashcroft	0.86	0.63	0.73	19
Junichiro Koizumi	0.94	0.83	0.88	18
Serena Williams	0.80	0.84	0.82	19
Tony Blair	0.66	0.71	0.68	38

Fig. 3. Experimental results

Step 2: according to the face recognition process based on PCA and SVM, PAC is used to reduce the dimension, and then SVM algorithm is used to recognize the face. The data is reduced to 100 dimensions, and the SVM parameters remain unchanged. The experimental results are as follows:

	precision	recall	f1-score	support
Ariel Sharon	0.89	0.81	0.85	21
Colin Powell	0.62	0.93	0.74	57
Donald Rumsfeld	0.83	0.76	0.79	33
George W Bush	0.77	0.89	0.82	111
Gerhard Schroeder	0.77	0.83	0.80	29
Hugo Chavez	0.93	0.67	0.78	21
Jacques Chirac	1.00	0.50	0.67	10
Jean Chretien	1.00	0.71	0.83	14
John Ashcroft	1.00	0.63	0.77	19
Junichiro Koizumi	1.00	0.61	0.76	18
Serena Williams	0.83	0.53	0.65	19
Tony Blair	0.83	0.66	0.74	38

Fig. 4. Experimental results

By comparing the two experimental results, it is found that the precision and recall performance indexes of face recognition have been significantly improved after PCA dimensionality reduction. The reason is that the dimensionality reduction not only reduces the computational complexity, but also removes some noise data, which is conducive to machine learning, and then improves the accuracy of face recognition.

Step 3: in order to better optimize the model, the parameters of the model need to be adjusted. The grid search parameter adjuster named gridsearchcv is used to adjust the parameters. The advantage of the grid search parameter adjuster is that it can automatically adjust the parameters and find the parameters with the best performance within a certain range. First, set the sequence of parameter C and gamma respectively, and the two parameters are randomly combined and adjusted according to the given order, Then compare the results of the given parameters and return the parameters with the best performance. The experimental results are as follows:

	precision	recall	f1-score	support
Ariel Sharon	0.94	0.75	0.83	20
Colin Powell	0.89	0.94	0.91	63
Donald Rumsfeld	0.80	0.74	0.77	27
George W Bush	0.81	0.95	0.87	134
Gerhard Schroeder	0.75	0.68	0.71	31
Hugo Chavez	0.75	0.69	0.72	13
Jacques Chirac	0.80	0.57	0.67	14
Jean Chretien	1.00	0.70	0.82	10
John Ashcroft	0.78	0.58	0.67	12
Junichiro Koizumi	1.00	0.72	0.84	18
Serena Williams	0.70	0.78	0.74	9
Tony Blair	0.76	0.72	0.74	39

Fig. 5. Experimental results

In the experiment, the sequence of C value and gamma value are set respectively, and the two parameters are randomly combined. The accuracy after parameter adjustment will be improved. The parameters can be further adjusted to improve the precision and recall.

Step 4: visualization

In order to further display the effect of face image recognition, the results of face prediction are visualized. The prediction of the second picture and the third picture in the first row in the figure is wrong, and the other pictures are correct. The experimental results show that the experimental effect of the face recognition method based on PCA and SVM is satisfactory, and it is feasible to carry out face recognition in a specific face database. The visualization diagram is shown in Figure 6:



Fig. 6. Experimental results

VII. CONCLUSION

Face recognition is one of the key research fields of artificial intelligence and pattern recognition. This paper discusses the problem of face recognition by combining SVM and PCA algorithm, and optimizes the important parameters, which not only reduces the amount of calculation and storage space of data processing, but also increases the accuracy of

recognition. The combination of the two has good feasibility and practical significance. The research on face recognition technology is conducive to further promote the wide application of face recognition.

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REFERENCES.

- [1] Zhi Hui,, Liu Sanyang. Face recognition based on genetic algorithm[J].Journal of Visual Communication and Image Representation, 2019,58: 495-502.
- [2] Jain N, Kumar S, Kumar A, et al. Hybrid deep neural networks for face emotion recognition[J].Pattern Recognition Letters, 2018, 115: 101-106.
- [3] Xu Jingze, Wu zuohong, Xu Yan. Face recognition based on PCA, LDA and SVM [J] Computer engineering and Applications , 2019,55(18) :34-37 (In Chinese)
- [4] HAN Dong, WANG Xuejun. Multi Pose Face recognition Based on Improved Convolutional Neural Network[J]. . Journal of Jilin University: Information Science Edition, 2018, 36(5) : 376-381 (In Chinese)
- [5] Reddy N , Rao M , Satyanarayana C . A Novel Face Recognition System by the Combination of Multiple Feature Descriptors[J]. The international arab journal of information technology, 2019, 16(4):669-676.
- [6] Xu X F , Zhang L , Duan C D , et al. Research on Inception Module Incorporated Siamese Convolutional Neural Networks to Realize Face Recognition[J]. IEEE Access, 2020, 8:12168-12178.
- [7] Utami N A , Maharani W , Atastina I . Personality Classification of Facebook Users According to Big Five Personality Using SVM (Support Vector Machine) Method[J]. Procedia Computer Science, 2021, 179(1):177-184.
- [8] Bashbaghi S , Granger E , Sabourin R , et al. Dynamic Ensembles of Exemplar-SVMs for Still-to-Video Face Recognition[J]. Pattern Recognition, 2017, 69:61-81.
- [9] Wen Y , Zhang K , Li Z , et al. A Comprehensive Study on Center Loss for Deep Face Recognition[J]. International Journal of Computer Vision, 2019, 127(6):668-683.
- [10] Zhang Y , Peng H . Sample reconstruction with deep autoencoder for one sample per person face recognition[J]. IET Computer Vision, 2017, 11(6):471-478.